

Game 06 — Flying Snake / Flappy (Complete Plan)

Game 6 of 7 Complexity: ★★ Medium Estimate: 1–2 days Status: Built

Complete plan (supersedes the 2026-09-09 sample). Status: **Built** (P0–P2 complete, 2026-09-11) — full trio in `apps/frontend/public/games/flying-snake/` (`core/render/main`), jest suite + `core.test.html` green; P3 extras (moving pipes, night palette, ghost run) deferred. Slug `flying-snake` ; folder `apps/frontend/public/games/flying-snake/` .
Branding decided: page title "Flying Snake" (master README §8 #8).
Committed 2026-09-11 (owner decision): the folder joined the committed games so its already-committed jest suite stays CI-safe (\$12 Rev 2 pass still pending).

1. Overview

Flappy-style: tap to flap against gravity, thread the gaps. One-tap, instantly restartable, brutally scoreable. Success metric: median session ≥ 8 runs ("one more try" loop confirmed); death always reads as fair.

2. Complete game spec

Physics (fixed timestep 1/120 s, accumulator, 250 ms frame clamp)

Constant	Value	Notes
GRAVITY	1800 px/s ²	
FLAP_VY	−480 px/s	one flap per press (no hold-to-fly)
TERMINAL_VY	+700 px/s	max fall speed
SNAKE_X	30 % width	fixed; world scrolls left
SNAKE_R	14 px	visual radius; hitbox radius ×0.9 (forgiving)
WORLD_SPEED	160 px/s	constant — difficulty scales via gap, not speed

Obstacles (pipe/vine pairs)

- **Gap:** 170 px, shrinks 2 px per point, floor 120 px.
- **Spacing:** 300 px horizontal, constant (density feels fairer than shrinking both axes).
- **Gap center:** random, ≥ 80 px + gap/2 from top and bottom edges.
- **Ceiling grace:** first 2 s of a run the ceiling clamps (no death); ground always lethal.
- **Collision:** circle (×0.9) vs pipe rects + ground strip; circle-rect standard clamp test.
- **Score:** +1 when the snake's x passes a pair's right edge (counted once per pair).
- **Medals:** bronze 10 · silver 20 · gold 40 · platinum 75 (gameover card).

Feel carriers

- **Tilt:** angle = `clamp(map(vy, -480...+700, -25°...+80°))` — the snake reads velocity.
- **Trailing segments:** 5 circles sampled from past positions every 40 ms with a small sinusoidal wiggle — sells "flying snake" with zero art assets.
- Death: white flash 120 ms + 300 ms screen shake + fall animation, then gameover card.

3. Screens & UI

State	Elements
<code>menu</code>	Title, best score + medal, bobbing snake animation, "Tap to start"
<code>ready</code>	World visible + frozen, "Tap to flap" hint; first tap starts physics mid-screen
<code>playing</code>	World + HUD score (large, top-center) + mute toggle
<code>paused</code>	Overlay (auto on <code>visibilitychange</code>)
<code>gameover</code>	Score, best, medal, [ Retry] [ Share]

Virtual viewport 720×960 logical (portrait), scaled/letterboxed, `devicePixelRatio` aware.

`?debug=1` draws hitbox circle + pipe rects (dev aid).

4. Controls

Tap anywhere / Space / ↑ = flap. `pointerdown` based; multi-touch extra flaps within 50 ms ignored; `touch-action:none`. No other input at MVP.

5. Data model (localStorage)

```
game:flying-snake:best  → { score: 23 }
game:flying-snake:prefs  → { muted: false }
```

6. File structure & function inventory

```
flying-snake/
  index.html
  style.css
  core.js    # pure: physics step, spawner, collision, scoring – test surface
  render.js  # procedural draw: snake segments, vines, ground, HUD
  main.js    # state machine (menu/ready/playing/paused/gameover) + glue
```

`core.js` exports: `createSnake()`, `stepSnake(snake, dt, flapped)` (gravity/terminal/flap), `createSpawner(rng)`, `nextGap(spawner, score)` (margins + shrink rule), `circleRectHit(cx, cy, r, rect)`, `tiltFor(vy)`, `medalFor(score)`. `main.js` owns the

run lifecycle: `ready` starts frozen; first flap starts physics; score-on-pass check in the update loop.

7. Edge cases

1. `ready` state: physics frozen, snake bobs at safe y (never inside a spawn line); first flap both starts the run and applies the flap.
2. Backgrounded → pause via shared loop; resume continues the exact fall state.
3. Flap during `gameover` → ignored; retry is a deliberate button (prevents skip-death).
4. Gap center RNG can't place a gap within margins (assert in spawner tests).
5. Score increments once per pair even if the snake weaves back (x only moves forward — snake x is fixed; world moves — so structurally impossible, test asserts it).
6. Storage disabled → no best/medal, playable.
7. Tab-titling: document.title swap on death is P3 fluff — skip.

8. Testing plan (core.js)

- Flap arc: from gap center y, a single flap clears the smallest gap (120 px) without a second flap when centered; two flaps suffice from gap edges (assert math).
- Spawner sweep: 1,000 gaps across scores 0–80 → all margins $\geq 80 \text{ px} + \text{gap}/2$; gap ≥ 120 .
- `circleRectHit`: corner cases (corner clamp), 1 px inside/outside inset behavior.
- `tiltFor`: endpoints (-25° at min vy, $+80^\circ$ at terminal), monotonic.
- `medalFor`: 9/10/19/20/39/40/74/75 boundaries.

9. Task breakdown

P0 — playable core

- ☒ `core.js` physics + spawner + circle-rect collision + score-once (+ tests §8)
- ☒ World rendering: pipes, ground, snake body with tilt + trailing segments
- ☒ menu/ready/playing/gameover states; ceiling grace; death flash/shake; instant retry
- ☒ Best score persistence; pause via `visibilitychange`

P1 — full rules & feel

- ☒ Gap shrink ramp; medals on gameover card; score pop animation
- ☒ WebAudio flap/score/hit + mute; menu bobbing animation

P2 — persistence/share/QA

- ☒ Share (`Flew through {n} gaps – beat that!`); `?debug=1` hitboxes
- ☒ QA gate (master README §5): fps check, offline, isolation greps (browser smoke test at 420×860 covered every state + persistence; a human 10-minute session is still owed before calling it player-proof)

P3 — polish

- ☐ Moving pipes (gentle bob) after score 50; night palette every 25 points; ghost of best run flying alongside `_(deferred)_`

10. Acceptance criteria

- §8 tests green; 1,000-gap sweep proof stored in the test file.
- Deterministic physics across 60/144 Hz (sim log comparison).
- First flap from `ready` never spawns the snake into a pipe.
- Restart ≤ 1 tap, < 300 ms to playable; 60 fps mid-range phone.

11. Deferred coupling (intentionally NOT built)

Footer/nav links, analytics events, achievements, leaderboards — per master README §7.

12. Rev 2 architecture upgrade (2026-09-11) — reference: `03-sliding-puzzle.md`

Owner-approved architecture reference for all games (Sliding Puzzle Rev 2), applied per this game's nature: flappy is score-attack with a seeded spawner, so async "same pipes" challenges are the natural multiplayer shape; the upgrade is persistence hygiene + those challenges. Isolation contract unchanged.

Target additions

```
flying-snake/
  index.html
  style.css
  core.js      # unchanged (physics/spawner/collision/scoring – test surface)
  render.js    # unchanged
  main.js      # unchanged
  config.js    # flags + per-locale strings (share template, medal copy), host-overridable
  storage.js   # guarded facade: versioned best/prefs + migrations, remote adapter slot
```

config.js

Physics constants stay in `core.js` — §2 tables are the spec of record. `config.js` carries flags and per-locale strings: share template, medal names/copy, per-locale UI copy. Host override `window.__FLYING_SNAKE_CONFIG__` → `?locale=` → defaults. `prefs` may override copy but is a cache, not the source of truth.

storage.js facade

Versioned save (`{ version, best, prefs:{muted} }`, migrated from the loose keys) with a remote adapter slot reserved for future account sync — the game never checks auth;

guests keep full local persistence. Mid-run resume: N/A (runs are seconds; death = instant restart by design).

Multiplayer (per this game's nature)

- **Async seeded pipes — the natural fit:** `createSpawner(rng)` already accepts a seed, so `?seed=` + a share template carrying it = the **identical pipe sequence** for a friend; compare score. Pairs naturally with the deferred P3 "ghost of best run".
- **Hot-seat:** alternate runs on one device, compare score — trivial.
- **Live race: gated** — backend + server-side timing + \$7 reversal. Not planned.

Phases

- ☒ R2-1 Hygiene (2026-09-11; strings wired 2026-09-12): `config.js` (locale + share/medal strings, host-overrideable) + `storage.js` (versioned `save` document — best + prefs — migrated from the loose best/prefs keys, corrupt-save sanitization, remote adapter slot); `main.js` slimmed to the facade. 42/42 tests green; flap-spam smoke clean.
- ☐ R2-2 Async seeded challenge: `?seed=` + share template carrying the seed — `createSpawner(rng)` is ready; `main.js` needs to thread the seed through.
- ☐ R2-3 A11y/polish: reduced-motion variant for flash/shake, menu/gameover focus handling, HUD contrast check.